AI-Assisted Systematic Reviews and Meta-Analysis

Day 2: From Corpus to Included Studies

Moses Boudourides

Faculty, Graduate Program on Data Science
Northwestern University

Moses.Boudourides@northwestern.edu

Moses.Boudourides@gmail.com

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Day 2: From Corpus to Included Studies

1. The Abstract Screening Bottleneck
2. Active Learning for Prioritization
3. LLM Zero-Shot Classification
4. Validation and Transparency
5. Practical Session and BYOD

The Screening Challenge

Day 1 ended with a deduplicated corpus of potentially thousands of records. The next phase in the systematic review lifecycle is screening titles and abstracts against predefined inclusion and exclusion criteria.

In a traditional manual workflow:

- Two independent reviewers read every abstract
- Disagreements are resolved by a third reviewer
- This process often takes weeks or months
- Reviewer fatigue leads to inconsistent application of criteria

AI as a Screening Assistant

The emergence of machine learning and natural language processing has created unprecedented opportunities to automate this labor-intensive stage.

However, the goal is **not** to replace human judgment with a fully automated, unsupervised black box. The goal is to use AI to prioritize relevant studies and apply criteria consistently, while maintaining human-in-the-loop oversight and rigorous validation.

What is Active Learning?

Active learning is a machine learning paradigm where the algorithm interactively queries a human user (or a set of rules) to label new data points. It is highly effective for systematic review screening (e.g., ASReview; Van de Schoot et al., 2021).

The core logic:

1. The model is initialized with a few known "relevant" and "irrelevant" papers.
2. The model trains a classifier (e.g., Logistic Regression or SVM) on the text features (TF-IDF) of those papers.
3. The model predicts the relevance of all unreviewed papers.
4. The model presents the most likely relevant papers to the reviewer.
5. As the reviewer labels them, the model retrains and updates its predictions.

Efficiency Gains from Active Learning

Simulation studies demonstrate that active learning can yield far more efficient reviewing than manual screening.

Researchers often identify 95% of relevant studies after screening only 5–10% of the corpus (Van de Schoot et al., 2021; Teijema et al., 2023).

In our workflow, we use a keyword-seeded Active Learning approach:

- We define "include" and "exclude" keywords.
- The model uses these to auto-label a training set.
- It then ranks the entire corpus by a Relevance Score (0 to 1).

Large Language Models in Screening

While Active Learning ranks papers by statistical relevance, Large Language Models (LLMs) like GPT-4 or Llama 3 can perform **zero-shot classification**.

Zero-shot means the model applies complex, plain-language inclusion/exclusion criteria to an abstract without requiring prior training examples.

Recent studies (Mahmoudi et al., 2025; Kim et al., 2025) show that properly prompted LLMs can achieve high sensitivity (recall) in abstract screening, often matching or exceeding human performance on complex criteria.

Prompt Engineering for Screening

The success of LLM screening depends entirely on the prompt. A rigorous screening prompt must:

- Define the role ("You are a systematic review screener.")
- Provide the exact inclusion and exclusion criteria.
- Provide the Title and Abstract.
- Demand a structured output (e.g., exactly "Include" or "Exclude").
- Demand a brief justification for the decision.

Demanding a justification is crucial: it forces the model to articulate its reasoning, allowing the human reviewer to spot-check for hallucination or misinterpretation of criteria.

The Necessity of Human Verification

AI screening is not infallible. Technical failure modes include:

- Scope creep (the LLM gradually misinterprets criteria over hundreds of prompts)
- Algorithmic bias (favoring certain methodologies or regions)
- Failure to handle unstructured or truncated abstracts

Therefore, human verification is non-negotiable. The standard validation protocol requires the human reviewer to manually screen a random sample (e.g., 10%) of the LLM's decisions and calculate inter-rater reliability (Cohen's Kappa).

The Transparency Log

To satisfy reporting standards (PRISMA 2020), every automated decision must be documented.

Our workflow generates a **Transparency Log** containing:

- The Title and Abstract
- The Active Learning Relevance Score
- The LLM Decision (Include/Exclude)
- The LLM Justification

This log allows researchers to audit the AI's logic and serves as the definitive record of the screening phase.

Hands-On: Screening the Corpus

In the second half of today's session, we move to the Streamlit app.

Guided Examples: We will run the Active Learning and LLM screening modules on the corpora generated on Day 1 (Health Inequalities, UBI, Microplastics, CSR). We will inspect the Transparency Logs and observe how the LLM justifies its exclusions.

Bring Your Own Data (BYOD): You will upload your Day 1 corpus, define your own inclusion/exclusion criteria in plain text, and run the Hugging Face-powered LLM screening live. You will download your final set of "Included Studies," ready for data extraction on Day 3.

References

- Boudourides, M. (2026). AI-Assisted Systematic Reviews and Meta-Analysis.
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- Teijema et al. (2023). Active learning using deep learning-based models for systematic reviews.
- Van de Schoot et al. (2021). An open source machine learning framework for efficient and transparent systematic reviews.

Questions and Discussion

Thank you!

Questions?

Moses.Boudourides@northwestern.edu
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